

Researchers: NASA Asteroid Samples Contain ‘Building Blocks’ of Life

Scientists examining material collected from an asteroid say it appears to contain some of the chemical building blocks of life.

The American spacecraft OSIRIS-REx collected the samples in 2020 from the asteroid Bennu. In 2023, the spacecraft sent the rock and dust material back to Earth in a special container that landed by parachute. Scientists from the American space agency NASA recovered the samples in the western state of Utah.

OSIRIS-REx is already on its way to another asteroid called Apophis. It is expected to reach that asteroid in 2029.

On Bennu, the spacecraft collected a total of 122 grams of material. This is believed to be the largest sample collected from beyond the moon. NASA has shared some of the material with international researchers.

Two research groups released studies last week describing results of their examinations of the asteroid material.

One study, published in *Nature Astronomy*, suggested the samples contained a mixture of organic compounds. Organic compounds have one or more carbon atoms that attach to other elements, usually hydrogen, oxygen, nitrogen and sulfur. All life on Earth is based on carbon and is built from organic compounds.

The other study, published in *Nature*, provides evidence that a much larger object, which Bennu is believed to have broken off from, once held salty, liquid water.

The researchers said this suggests that in the early solar system, asteroids such as Bennu might have transported water and life-supporting chemicals to other planets and moons.

Researchers from both groups noted it was important that the samples they worked with came directly from the asteroid itself. Asteroid pieces that fall to Earth turn into meteorites, and chemicals inside the samples can be changed or lost.

In a statement, NASA said the findings do not provide direct evidence of life itself. However, the space agency said, “They do suggest the conditions necessary for the emergence of life were widespread across the early solar system.” This increases the chances that “life could have formed on other planets and moons,” the statement added.

The parent of Bennu is believed to be an icy body measuring about 100 kilometers across. Scientists believe the body formed in the outer solar system and was later destroyed, possibly 1 to 2 billion years ago. The pieces that broke off likely formed Bennu and other asteroids observed to be masses of loose material, rather than solid objects.

Nicky Fox is the associate administrator for NASA's Science Mission Directorate in Washington D.C. She praised the OSIRIS-REx mission, saying it was "already rewriting the textbook on what we understand about the beginnings of our solar system."

Fox added that she hopes the samples can help scientists understand "what ingredients in our solar system existed before life started on Earth."

Tim McCoy is a researcher at the U.S. Smithsonian Institution and was a lead writer of the Nature study. He told the Associated Press (AP) that the samples may provide details about "the kind of environment that could have been essential to the steps that lead from elements to life."

He noted that combining the ingredients of life with an environment of sodium-rich water is "really the pathway to life." He added the samples suggest the processes involving the chemical compounds "probably occurred much earlier and were much more widespread than we had thought before."

Yasuhito Sekine is with the Institute of Science in Tokyo. He did not take part in the new research. Sekine confirmed the findings were only made possible by examining samples "collected directly from the asteroid, then carefully preserved back on Earth."

Jason Dworkin is a project scientist for the OSIRIS-REx mission at NASA's Goddard Space Flight Center in Maryland. He told Reuters news agency that all biology is made up of organic compounds. And some of those may have survived 4.5 billion years.

Dworkin said the materials from OSIRIS-REx provide new details about a solar system that might have had the right elements to support life. But he added that one of the most interesting questions, still remains: "Why we, so far, only see life on Earth and not elsewhere."

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from The Associated Press, Reuters, Agence France-Presse and NASA.

Quiz - Researchers: NASA Asteroid Samples Contain 'Building Blocks' of Life

Vocabulary

asteroid - n. ดาวเคราะห์น้อย (วัตถุที่ประกอบด้วยหินและโลหะที่โคจรรอบดวงอาทิตย์ แต่มีขนาดเล็กกว่าดาวเคราะห์)

sample - n. ตัวอย่าง (ปริมาณเล็กน้อยของสิ่งของที่ให้ข้อมูลเกี่ยวกับสิ่งนั้นๆ)

organic - adj. อินทรีย์ (ที่มาจากสิ่งมีชีวิต)

emerge - v. ปรากฏขึ้น (โผล่ออกมาจากที่ใดที่หนึ่ง)

loose - adj. หลวม (ไม่ได้ยึดติดแน่น)

ingredient - n. ส่วนประกอบ (ชิ้นส่วนต่างๆ ที่รวมกันเป็นสิ่งใดสิ่งหนึ่ง)

preserve - v. รักษาสภาพ (ทำให้สิ่งใดสิ่งหนึ่งคงเดิมหรือป้องกันไม่ให้เสียหาย)

Reading Comprehension Quiz

1. What do scientists believe the material collected from an asteroid contains?

- A) Signs of extraterrestrial intelligence
- B) The chemical building blocks of life
- C) Solid evidence of ancient alien civilizations
- D) Unusual geological formations

2. Which American spacecraft collected samples from the asteroid Bennu?

- A) Voyager 1
- B) Mars Rover
- C) OSIRIS-REx
- D) Hubble Space Telescope

3. When were the samples collected by OSIRIS-REx sent back to Earth?

- A) In 2020
- B) In 2023
- C) In 2029
- D) Not yet returned

4. What is the next asteroid OSIRIS-REx is heading towards after Bennu?

- A) Vesta
- B) Ceres
- C) Apophis
- D) Eros

5. How much material was collected by OSIRIS-REx from Bennu?

- A) 100 kilograms
- B) 122 grams

- C) 200 grams
- D) 10 kilograms

6. What makes the sample collected from Bennu particularly significant, according to the article?

- A) It is the first sample ever collected from an asteroid.
- B) It contains rare elements not found on Earth.
- C) It is believed to be the largest sample collected from beyond the moon.
- D) It was collected by an international collaboration of space agencies.

7. How many research groups publicly reported their findings from examining the asteroid material last week?

- A) One
- B) Two
- C) Three
- D) Several

8. According to one study published in Nature Astronomy, what did the Bennu samples contain?

- A) Traces of ancient microbial life
- B) A mixture of organic compounds
- C) Pure carbon crystals
- D) High concentrations of heavy metals

9. The other study, published in Nature, provided evidence that Bennu's parent object once contained what?

- A) A molten core
- B) An atmosphere rich in oxygen
- C) Salty, liquid water
- D) Large deposits of ice

10. What do researchers suggest about asteroids like Bennu in the early solar system?

- A) They were primarily responsible for planetary collisions.
- B) They were too small to influence planetary development.
- C) They might have transported water and life-supporting chemicals to other planets.
- D) They primarily orbited gas giants, not inner planets.

11. Why did researchers emphasize the importance of samples coming directly from the asteroid, rather than meteorites?

- A) Asteroids are easier to access than meteorites.
- B) Meteorites are too small to provide useful data.
- C) Chemicals in meteorites can be changed or lost after falling to Earth.
- D) Asteroids provide a greater variety of geological formations.

12. What does NASA explicitly state the findings do NOT provide direct evidence for?

- A) The existence of organic compounds.
- B) The presence of water in the early solar system.
- C) Life itself.
- D) The formation of asteroids.

13. What is believed to be the nature of Bennu's parent body?

- A) A dense, metallic object
- B) An icy body about 100 kilometers across
- C) A volcanic planetoid
- D) A comet from the Oort Cloud

14. How did Nicky Fox, NASA's associate administrator, describe the OSIRIS-REx mission?

- A) A challenging but ultimately unsuccessful endeavor.
- B) A mission that has surpassed all previous space explorations.
- C) A mission 'already rewriting the textbook' on understanding the solar system's beginnings.
- D) A preliminary step before more ambitious missions.

15. According to Jason Dworkin, what is still one of the most interesting unanswered questions regarding life in the solar system?

- A) How organic compounds can survive for billions of years.
- B) The exact chemical processes leading to life's emergence.
- C) Why life is only observed on Earth and not elsewhere.
- D) Whether other planets also have salty, liquid water.